

PC-SAFT in the packing fraction: separating temperature and density

Internal working note (derivation record; not manuscript text)

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Scope: PC-SAFT hard chain + dispersion (Gross & Sadowski 2001), non-associating, non-polar, arbitrary number of components, constant k_{ij} . Every identity below was checked numerically against a direct implementation (Sec. 6).

1 Why a (T, ρ) subdivision looks impossible

In (T, ρ) the hard-sphere pole sits at $\rho = 1/q(T)$, with q defined in Eq. (2). Because the segment diameter $d_i(T)$ depends on temperature, the pole moves with T . A fixed rectangle in (T, ρ) therefore either contains the pole for some T or wastes most of its area. The mixing rules are also not mole-fraction-weighted sums of per-component terms. The resolution is to use η rather than ρ as the density coordinate. After that, temperature needs *no* subdivision at all.

2 Notation

$$d_i(T) = \sigma_i \left[1 - 0.12 e^{-3\varepsilon_i/(kT)} \right], \quad s_n(T, \mathbf{x}) = \sum_i x_i m_i d_i^n \quad (n = 0 \dots 3), \quad \bar{m} = \sum_i x_i m_i, \quad (1)$$

$$r_n = s_n/s_3, \quad q(T, \mathbf{x}) = \frac{\pi}{6} N_A 10^{-30} s_3, \quad \eta = \zeta_3 = \rho q. \quad (2)$$

Here σ is in Å, ρ in mol m⁻³ and q in m³ mol⁻¹. The substitution that drives everything is

$$\boxed{\zeta_n = \frac{\pi}{6} \rho N_A 10^{-30} s_n = \eta r_n(T, \mathbf{x})} \quad (3)$$

All ζ_n are proportional to η , with coefficients that depend only on $(T, \mathbf{x})$. The number density follows as $\rho N_A 10^{-30} = 6\eta/(\pi s_3)$.

3 Term by term

Hard sphere. Substituting $\zeta_n = \eta r_n$ into the Boublík–Mansoori form, every η -dependence collapses onto three universal functions:

$$a^{\text{hs}} = A_1 \frac{\eta}{1 - \eta} + A_2 \frac{\eta}{(1 - \eta)^2} + A_3 \ln(1 - \eta), \quad A_1 = \frac{3r_1 r_2}{r_0}, \quad A_2 = \frac{r_2^3}{r_0}, \quad A_3 = \frac{r_2^3}{r_0} - 1. \quad (4)$$

Chain. With $D_i = d_i/2$, define

$$a_i(T, \mathbf{x}) = 3D_i r_2. \quad (5)$$

The contact value g_{ii}^{hs} then depends on $(T, \mathbf{x})$ only through a_i , because the coefficient of the third term is $2D_i^2 r_2^2 = 2a_i^2/9$ identically. Its numerator factorizes exactly:

$$g_{ii}^{\text{hs}} = \frac{(1-\eta)^2 + a_i\eta(1-\eta) + \frac{2a_i^2}{9}\eta^2}{(1-\eta)^3} = \frac{(1 - \frac{3-a_i}{3}\eta)(1 + \frac{2a_i-3}{3}\eta)}{(1-\eta)^3}, \quad (6)$$

$$\ln g_{ii}^{\text{hs}} = \ln(1 - \frac{3-a_i}{3}\eta) + \ln(1 + \frac{2a_i-3}{3}\eta) - 3\ln(1-\eta). \quad (7)$$

For a pure fluid, $a = 3/2$ and this reduces to Carnahan–Starling, $g = (1 - \eta/2)/(1 - \eta)^3$. The branch points are at $\eta = 3/(3 - a_i)$ and $\eta = -3/(2a_i - 3)$. Both satisfy $|\eta| \geq 1$ for $0 \leq a_i \leq 3$, i.e. for diameter ratios up to 2.

Dispersion. The double sums are T -free up to explicit powers of $1/T$:

$$\overline{m^2 \varepsilon \sigma^3} = \frac{E_1}{T}, \quad \overline{m^2 \varepsilon^2 \sigma^3} = \frac{E_2}{T^2}, \quad E_\nu = \sum_{i,j} x_i x_j m_i m_j \sigma_{ij}^3 \left[\frac{\varepsilon_{ij}}{k} \right]^\nu, \quad \varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j} (1 - k_{ij}). \quad (8)$$

Using $\rho N_A 10^{-30} = 6\eta/(\pi s_3)$:

$$a^{\text{disp}} = B_1 \eta I_1(\eta; \bar{m}) + B_2 \eta C_1(\eta; \bar{m}) I_2(\eta; \bar{m}), \quad B_1 = -\frac{12E_1}{s_3 T}, \quad B_2 = -\frac{6\bar{m}E_2}{s_3 T^2}. \quad (9)$$

I_1 and I_2 are affine in $(c_1, c_2) = (\frac{\bar{m}-1}{\bar{m}}, \frac{\bar{m}-1}{\bar{m}} \frac{\bar{m}-2}{\bar{m}})$, e.g. $I_1 = \sum_{k=0}^6 (a_{0k} + c_1 a_{1k} + c_2 a_{2k}) \eta^k$.

4 Result

At fixed $(T, \mathbf{x})$ the residual Helmholtz energy is a function of η alone:

$$\alpha^r(T, \rho, \mathbf{x}) = \sum_k c_k(T, \mathbf{x}) \phi_k(\eta) - \sum_i w_i \ln g^{\text{hs}}(\eta; a_i(T, \mathbf{x})), \quad \eta = \rho q(T, \mathbf{x}) \quad (10)$$

weight $c_k(T, \mathbf{x})$	basis $\phi_k(\eta)$	depends on
$\bar{m}A_1$	$\eta/(1-\eta)$	universal
$\bar{m}A_2$	$\eta/(1-\eta)^2$	universal
$\bar{m}A_3$	$\ln(1-\eta)$	universal
$B_1, B_1 c_1, B_1 c_2$	$\eta \sum_k a_{0k} \eta^k, \eta \sum_k a_{1k} \eta^k, \eta \sum_k a_{2k} \eta^k$	universal
B_2	$\eta C_1(\eta; \bar{m}) I_2(\eta; \bar{m})$	one parameter, $\bar{m}$
$w_i = x_i(m_i - 1)$	$\ln g^{\text{hs}}(\eta; a_i)$	one parameter, a_i

The weights are cheap scalars: N exponentials for $d_i(T)$ plus sums.

Pressure. Since $\rho \partial_\rho|_{T, \mathbf{x}} = \eta \partial_\eta|_{T, \mathbf{x}}$, we have $Z = 1 + \eta \partial \alpha^r / \partial \eta$. Explicitly,

$$Z = 1 + \bar{m} \left[A_1 \frac{\eta}{(1-\eta)^2} + A_2 \frac{\eta(1+\eta)}{(1-\eta)^3} - A_3 \frac{\eta}{1-\eta} \right] + B_1 \eta \frac{d(\eta I_1)}{d\eta} + B_2 \eta \frac{d(\eta C_1 I_2)}{d\eta} - \sum_i w_i \eta \left[\frac{3}{1-\eta} - \frac{(3-a_i)/3}{1 - \frac{3-a_i}{3}\eta} + \frac{(2a_i-3)/3}{1 + \frac{2a_i-3}{3}\eta} \right], \quad (11)$$

and $p = \rho RTZ$ becomes

$$F(\eta) \equiv \eta Z(\eta) = \frac{pq(T, \mathbf{x})}{RT} \quad (12)$$

Pressure enters only as a constant on the right-hand side.

5 Consequence: subdivide in η only

Domain. $\eta \in [0, \eta_{\text{cp}}]$ with $\eta_{\text{cp}} = \pi/(3\sqrt{2}) \approx 0.7405$ (close packing). The interval is the same for every fluid, mixture, temperature and pressure. There is no compactification, and no calls fall outside the domain.

Singularities. These lie off the real interval and set the Chebyshev degree per piece:

source	location	remedy
$1/(1-\eta), \ln(1-\eta)$	$\eta = 1$	grade pieces toward η_{cp}
$\ln g^{\text{hs}}$	$\eta = 3/(3-a_i), -3/(2a_i-3), \eta \geq 1$	none needed
C_1	zeros of $P(\eta; \bar{m})$, real $\approx -1/(3\bar{m})$ for $\bar{m} \gtrsim 4$	multiply by P^2

Here $1/C_1 = P(\eta; \bar{m})/[(1-\eta)^4(2-\eta)^2]$, with

$$P = (1-\eta)^4(2-\eta)^2 + \bar{m}(8\eta - 2\eta^2)(2-\eta)^2 + (1-\bar{m})(20\eta - 27\eta^2 + 12\eta^3 - 2\eta^4)(1-\eta)^2, \quad (13)$$

a degree-6 polynomial that is *affine in $\bar{m}$* . For small η , $P = 4 + 12\bar{m}\eta + O(\eta^2)$. For long chains its zero therefore approaches $\eta = 0$ from the left, e.g. -0.011 at $\bar{m} = 30$, and would force many pieces. F contains C_1 and C'_1 , so P^2F is free of these poles.

Rootfinding function. On each piece, solve

$$G(\eta) = P^2[F(\eta) - t] = \eta Q(\eta) - tP^2(\eta), \quad Q = P^2Z, \quad t = \frac{pq}{RT}. \quad (14)$$

- Q is $O(1)$ and is what gets fitted.
- The factor η is applied exactly, as a Chebyshev multiply-by- x , so the absolute error near $\eta = 0$ scales with η and the gas root keeps full relative precision.
- P^2 has degree 12 and is represented exactly.
- t enters linearly, so one set of expansions per $(T, \mathbf{x})$ serves every pressure.

Towards universal tables. Every term of P^2F is a scalar weight in $(T, \mathbf{x})$ times a function of η that is *independent of the fluid*. Two families carry a parameter:

- the C_1I_2 term, through $\bar{m}$. P^2 is quadratic in $\bar{m}$, so it splits into three universal pieces;
- the chain term, through a_i : one 2-D table on a rectangle in (η, a) .

This is the stacked per-term representation of Bell & Alpert (2018), on fixed rectangles. It has not been built.

6 Numerical checks

Scripts are `pcsaft_cheb.py` and `spurious_scan.py` in `dev/autodiff_spike/`, on branch `claude/coolprop-autodiff-comparison-ff35be`.

check	result
factorization of the g^{hs} numerator	exact to 4×10^{-16}
explicit Z above vs. direct model (3 fluids, 150–600 K)	$\leq 1.3 \times 10^{-14}$ relative
root counts vs. brute-force scan (288 cases, 40–800 K, 10^{-4} – 10^3 MPa)	all match
relative root error, ~ 10 pieces at degree 12	$\leq 2 \times 10^{-9}$
pieces at degree 12 without / with the P^2 multiplier ($\bar{m} = 30$)	14 / 5 (singularity bound)

The singularity bound in the last row ignores how much Z varies in magnitude within a piece. About twice as many pieces are needed in practice.

Spurious roots. With the Gross–Sadowski constants, a fourth root appears at $\eta \approx 0.56$ – 0.74 (cf. Privat et al. 2010). It occurs only at or below the pure fluid’s triple point: methane ≤ 28 K, propane ≤ 84 K, n-decane ≤ 140 K. With the Liang 2012 or 2014 universal constants it does not occur at 20–400 K. That check paired the Liang constants with the Gross–Sadowski pure-fluid parameters, so it is qualitative only.

7 Not covered

- **Association.** The site fractions solve an implicit equation. At fixed $(T, \mathbf{x})$ the term is still smooth in η , so per- $(T, \mathbf{x})$ fits from node values work, but those tables are not universal.
- **Polar terms (Gross–Vrabec).** Their Padé form $a_2/(1 - a_3/a_2)$ has poles of its own, which need the same analysis as C_1 .
- **Temperature-dependent k_{ij}** only makes E_ν depend on T . That is harmless.